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Manipulation of Procurement Contracts: Evidence from the Introduction of Discretionary Thresholds[†]

By JÁN PALGUTA AND FILIP PERTOLD*

We present evidence of how policies that create opportunities to avoid open competition in procurement lead to the manipulation of procurement values. We exploit a policy reform in which public bodies were given the autonomy to preselect potential contractors below newly defined discretionary thresholds. Manipulation is revealed through bunching of procurements just below the thresholds in construction works and services, and to a lesser degree, in goods. Among manipulated contracts, we document a threefold increase in the probability that procurements are allocated to anonymous firms, which can hide their owners. This sorting violates assumptions behind regression-discontinuity designs. (JEL D73, D86, H57, K23, L74)

Administrative law has long recognized that giving greater discretion to public officials can adversely affect public service provision. Public servants can abuse discretionary power to pursue their private rather than social interests. However, a potential benefit of the discretion is that competent bureaucrats can use valuable information they hold privately about the provision of public services.

In the spirit of administrative law, many European countries, as well as the United States, use open auctions in public procurement. Open auctions are supposed to increase transparency, boost competition, and promote equal access among potential contractors, but they also reduce public officials' flexibility and generate additional administrative and time costs (Banfield 1975, Kelman 1990, Bulow and Klemperer 1996, Europe Economics 2006, Calzolari and Spagnolo 2009, and Helper and Henderson 2014). If the anticipated value of a procurement does not exceed a certain legislative threshold, public officials can decide to allocate procurements using

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flexible contract-allocation procedures in which firms selected by the officials submit bids.¹ This discretion enables knowledgeable public officials to exclude suppliers with potentially low performance, but it also increases the risk that dishonest officials will collude with some suppliers and ultimately misallocate public resources.

In this paper we ask whether and how legislative thresholds allowing for such flexible procedures affect public officials' decisions regarding the value of a given procurement and the selection of contractors. Understanding the effects of these thresholds on public officials' behavior is important, since OECD countries alone redistribute 13 percent of their GDP through public procurement systems (OECD 2013). At the same time, 40 percent of the total value in European construction works procurement is below the threshold level (European Commission 2011). Expenditures under the US simplified acquisition threshold (excluding micro-purchases) totaled about US\$19 billion in 2014, which is about 4 percent of federal acquisition spending (*GovConChannel* 2015).

Previous literature used legislative thresholds in regression-discontinuity designs in order to analyze whether contract-allocation procedures with different levels of discretion affect procurement outcomes, such as contract prices or renegotiation costs (Spagnolo 2012; Coviello and Mariniello 2014; and Coviello, Guglielmo, and Spagnolo 2016). Discontinuity in discretion at thresholds helped to address the identification problem that public officials may choose the type of allocation procedure based on unobserved characteristics of procurements, such as their complexity. In order to properly apply such discontinuity designs, these studies had to analyze only procurements that are not subject to sorting by public officials below the thresholds. However, little attention has been paid to procurement sectors where legislative thresholds may lead to strategic considerations of public officials—for example, in setting the anticipated value of procurement or using discretion to select specific contractors. Understanding the strategic behavior of public officials is critical for setting such thresholds in legislation as well as in discussion about optimal levels of discretion in the public sector.

We contribute to the literature by showing how the introduction of thresholds that allow the option to avoid open procurement competition leads to distortions by public officials. Distortions are revealed by bunching of procurement values just below the thresholds, and are more likely to be present in certain procurement sectors where setting the anticipated value is less transparent, such as in construction and services. We thus show that the contract-allocation procedure and the anticipated value of a procurement can be simultaneously subject to the choices of public officials. This implies that the discretion of public officials can endogenize the relationship between characteristics of procured goods, format of procurement (e.g., transparency, flexibility), and procurement outcomes (e.g., contract price).

We further show that manipulation of procurement values is associated with an increased probability that procurements are awarded to firms hiding their owners.

¹ The US Federal Acquisition Regulation (FAR), for example, has a "simplified acquisition threshold" set at a contract value of \$150,000. Several reporting requirements do not apply below this value; e.g., the Miller Act (requiring payment and performance bonds), etc. (Coviello, Guglielmo, and Spagnolo 2016). The threshold is especially significant for small businesses, because transactions below the threshold are reserved for small businesses.

We also observe that the contract prices (relative to the anticipated value of procurements) just below the thresholds are higher if procurements are awarded to anonymously owned firms rather than to firms with traceable owners.

We view our results as a concrete illustration of how endogenous sorting below discretionary thresholds may invalidate regression discontinuity designs, because characteristics of procurement winners and related procurement outcomes around the discretionary thresholds can be subject to the choices of public officials. Further, the observed association between the manipulation of procurement values and the revealed preferences for anonymous firm owners suggests that in a regime with higher flexibility, public officials are more likely to select firms whose legal form allows for hiding rent-seeking behavior.² This evidence is more in line with a traditional view of administrative law that emphasizes the costs of administrative discretion over its benefits, which may be relevant mainly for countries with low accountability of public officials and weak institutions, represented here by the case of the Czech Republic (Shleifer and Vishny 1993, 1994).

In our empirical strategy, we employ a policy change that introduced new discretionary thresholds into the system of public procurements. Before the change, public procurements had to be conducted through transparent auctions with open access for any potential contractor. After the change, public bodies were given the autonomy to preselect at least five potential contractors for simplified negotiations if the anticipated value of procurement was set below the legislative threshold. For example, construction procurements could be awarded using the new simplified procedure only if they were valued at less than CZK 20 million (approximately US\$1 million) (see Table A1 in the Appendix). By comparing the empirical distributions of procurement values before and after the reform, we can isolate the effect of the thresholds from other factors that may cause bunching in procurement and which are constant over time.

This approach has rarely been used in the economic literature, which usually analyzes the effect of thresholds on sorting in environments with already existing thresholds. For example, Urquiola and Verhoogen (2009) show that class size caps lead to endogenous sorting of students with specific parental backgrounds into schools near the cap. This illustrates how estimating the impact of class size on student output may lead to bias even in a regression discontinuity design. Another example is Saez (2010), which, in line with a simple tax evasion model, shows that tax brackets lead to sorting of the self-employed just below the bracket. In both cases, the distribution of the studied outcomes is not observable for environments without the thresholds, and the shape of the counterfactual needs to be assumed; for example, as smooth at the threshold. In many cases, such assumptions may be questioned, as in our case due to the rounding of procurement values by public officials. We thus incorporate

²In countries with weak institutions, anonymously owned firms may hide the identity of agents who are procurement officials and at the same time stakeholders in anonymously owned companies (Gordon 2009, Sharman 2010, Office for Drug Control and Crime Prevention 1998). Previous evidence from Russia, for example, shows that similar fly-by-night firms are used to extract resources from public procurement (Mironov and Zhuravskaya 2016). In Table A3 in the online Appendix we present anecdotal evidence of many corruption scandals that are associated with anonymously owned firms in public procurement in the Czech Republic.

the pre-reform distribution of procurement values into our empirical strategy. This approach ultimately leads to lower estimates of sorting.

In our analysis, we use data that contain information about more than 45,000 procurements worth over US\$52 billion, and include details about procurement contracts, winning contractors, and procurement authorities. This data originates from the Czech Republic, where the procurement market corresponds to approximately 16 percent of GDP and where cross-country comparisons indicate high levels of favoritism in public contracting (Transparency International 2012, Schwab 2011). The shortcoming of these data is that they do not include information about the delivered quality of procurements, renegotiations, and administrative costs, which precludes an overall welfare analysis of the impact of manipulation. Our dataset, however, does carry information about the characteristics of the winning firms, including information about the traceability of their owners.

Our research is in line with literature that discusses optimal delegation in the public sector. This literature (e.g., Alonso and Matouschek 2008) advises the use of stricter rules and external controls whenever there is a misalignment between the preferences of public officials and those of the society. This is supported by the empirical evidence presented in Di Tella and Schargrodsky (2003), which shows that external controls and reduced discretion of public officials led to less corruption in the procurement of medical goods in Argentina. In contrast, Bandiera, Prat, and Valletti (2009) argue that high external controls and reduced discretion can have detrimental effects when wasteful behavior is unintentional and public officials do not earn private benefits from public waste. Similarly, Spagnolo (2012); and Coviello, Guglielmo, and Spagnolo (2016) use a regression discontinuity design to document that greater use of flexible procedures below thresholds does not adversely affect outcomes in procurement, but allows public officials to speed up the procurement process by selecting incumbent suppliers. On the other hand, Coviello and Mariniello (2014) analyze the impact of reduced transparency requirements below procurement thresholds and show that reduced transparency has a negative impact.

In our case, we find that the introduction of regimes giving greater discretion to procurement officials in a high corruption country leads to the manipulation of procurement values and simultaneously to a more likely allocation of procurements just below the thresholds to firms with anonymous ownership structure. Although we cannot quantify the impact of the manipulation on quality of the provided goods or administrative and renegotiation costs, our results are more in line with the stream of literature suggesting likely negative effects of the reduced transparency, greater discretion, and low accountability of public officials.

The rest of the paper is organized as follows. Section I discusses the institutional framework of Czech public procurement and the policy reform that introduced new procurement thresholds. Section II describes the data from procurement contracts. Section III presents our empirical strategies for detecting manipulation in procurement. Section IV shows the results and the empirical analysis of manipulation, along with robustness checks. Section V presents evidence that the thresholds distort associations between the characteristics of contract-allocation procedures and procurement outcomes. Finally, Section VI summarizes our findings and discusses policy implications.

I. Institutional Background

A. Public Procurement and Thresholds in the Anticipated Value

Public procurement constitutes one of the largest public spending processes in the Czech Republic. Annually, about 13–16 percent of GDP (US\$31 billion in 2010) is spent on procurement of goods, construction works, and services, making this one of the largest procurement markets among OECD countries (OECD 2013).

Several characteristics of planned procurements determine the level of autonomy of procurement agencies in the Czech institutional setting. In this study, we focus on the *anticipated value* of procurements, as this defines the level of discretion available to procurement officials.

Different legislative thresholds in the anticipated contract value divide procurements into separate groups that differ in their mandatory requirements on transparency and open access to the procurement process. The rule is that the anticipated value should be set so as to approximate the anticipated financial obligations ensuing from the contract, so that the agency can plan its future expenditures. Engineers employed by the agencies estimate the value of projects prior to the start of the contract-awarding process based on a menu of standardized costs for each type of work. However, these standards vary across agencies and are neither strictly binding nor enforced. The anticipated value does not serve as a reserve price in procurement auctions. The firms can bid a higher price than the anticipated value of a project. As shown later, procuring agencies can thus adjust the anticipated value of projects quite freely, so that lighter legislative restrictions apply to their targeted procurements.

The reform of the Czech procurement code in July 2006 introduced a new type of procurement procedure into the legislation and thereby introduced several new thresholds in procurement (see Table A1 in the Appendix). This reform followed the accession of the Czech Republic into the European Union and replaced the older regulation which did not comply with the European directives on public procurement (Pavel 2013). The introduction of the new thresholds is the key factor for our empirical strategy, because the new procedure is not applicable above these thresholds.

The new thresholds offered procurement officials the opportunity to free themselves from the rigid rules which otherwise regulate the transparent contract-awarding process. In particular, if the anticipated value of a procurement contract is set below the threshold, the officials are allowed to autonomously approach potential contractors themselves instead of being required to provide open access to any company that might want to participate in the procurement competition. Agencies therefore did not need to set lengthy deadlines for bid submission or to evaluate all incoming bids, but could rather directly invite preselected companies to submit bids. In a trade-off, the law demanded that agencies must invite at least five potential suppliers so as to guarantee some degree of competition (see Table A2 in the Appendix for further details on procurement procedures).

A major novelty of the reform arose from the fact that the decision as to which bidders would be invited was left at the full discretion of the procurement agencies.

This practice is considered common in private sector procurement (Kelman 1990). The discretion to preselect subcontractors enables firms to procure contracts based on long-term relationships and reputation, which can partly remedy the common aspect of non-contractible quality in procurement (Bannerjee and Duflo 2000, Calzolari and Spagnolo 2009). The cost of this discretion is, however, less equal treatment of suppliers and a lower accountability of officials in less transparent procurement. In private sector procurement, these issues are nonetheless considered secondary, because private procurement officials do not redistribute public money and are accountable to company owners.

B. Anonymously Owned Firms in the Czech Republic

Czech public procurement has been criticized for favoritism, corruption, and lack of effective institutional oversight. Schwab (2011) ranked the Czech Republic as low as 123rd among 142 countries in terms of the extent to which government officials show favoritism toward well-connected firms. Even though two public institutions oversee Czech public procurement, one of them, the Supreme Audit Office, does not have the authority to impose sanctions, but merely issues recommendations, while the other, the Czech Antitrust Office, has been known for its rather passive and formalistic approach (Transparency International 2009).

The Czech institutional setting has also been characterized by a high number of anonymously owned firms competing for procurements. The Czech legal code specifically enables joint-stock companies to issue two types of shares: either they are nominated to concrete holders with shareholders' names declared directly on the shares (or in the list of shareholders), or a joint-stock company can issue bearer shares, which entitle any current bearer of the shares to property rights. The share bearers are not registered anywhere and are unknown to the joint-stock company or to any controlling bodies (Šedová 2014). A change in ownership can be performed instantly without producing any traceable records. The owners of bearer shares usually cash in benefits from their ownership by sending legal representatives to general meetings of joint-stock companies. The loose regulation of bearer shares thereby helps to conceal connections between businesses and procurement officials.

Many firms with anonymous owners have played an important role in various corruption scandals in both the public and private sector (*Economist* 2013), which is also documented in Table A3 in the online Appendix. These scandals have led to pressure from many anticorruption organizations, including Transparency International, demanding that anonymous ownership be banned in order to reduce corruption and other types of waste of public resources. The Czech government has incorporated these recommendations into legislation.

II. Data from Public Procurement Contracts

The available data on public procurement contracts include the characteristics of all procurements awarded in the Czech Republic from 2005 to 2010, conditional on their procurement process being governed by the Czech Public Procurement Act. The database therefore mandatorily contains information on contracts that are

TABLE 1—DESCRIPTIVE STATISTICS

Characterization	January 2005–June 2006		July 2006–Dec 2010	
	Volume (CZK billion)	Percent	Volume (CZK billion)	Percent
By main object				
Goods	21.56	12.45	102.71	11.87
Services	37.87	21.87	192.17	22.21
Construction works	113.71	65.68	570.33	65.92
By procurement procedure				
Open	146.52	84.58	580.45	68.00
Restricted	26.69	15.42	82.23	9.63
Simplified negotiations or negotiations with prior public notice	N/A	N/A	98.56	11.55
Negotiations without prior public notice	N/A	N/A	92.36	10.82
By procuring agency type				
National procurers	117.89	68.09	560.63	64.80
Regional public bodies	55.25	31.91	304.57	35.20
By anonymity of suppliers				
Anonymous owner	3.26	1.88	18.78	2.17

Notes: Descriptive statistics are provided in billion CZK (20 CZK ≈ 1 USD). The N/A mark indicates the non-applicability of a statistic for a given observation period.

expected to cost more than minimum anticipated value and does not contain data on contracts procured through legislative exemptions.³ Altogether, this amounts to over 45,000 procurement contracts worth more than CZK 1,038 billion (approximately US\$52 billion).

The unit of observation in this study is a procurement project, although several contracts (with different contractors) may be procured within one project, for example, due to geographically dispersed branches of some procurement agencies or due to procurement through framework agreements. The focus is however placed on projects because the anticipated value of procurements must be estimated at the level of an entire project, rather than for each contract separately.

For each project, the database includes information on the subject of the procured goods/services/construction works (represented by a detailed CPV code⁴), the type of contract-awarding procedure, characteristics of the procuring agencies and winning contractors, and the anticipated value and contract price of procurements.

Table 1 provides summary statistics of all contracts in two observation periods: before and after the legislative reform of the Public Procurement Act in 2006. The table shows very little change in the overall structure of demand for procurements. One can see that the share of the overall procurement value spent on goods, services, and construction remains stable over time. Also, there is only a small change in the share of procurements allocated by national versus regional agencies.

³The minimum anticipated value for collecting data on procurements was CZK 2 million for goods, services, and construction works before the 2006 reform. This was far below the new discretionary thresholds. The limit for construction works changed in 2006 from CZK 2 million to CZK 6 million. We take this change into account in our analysis.

⁴CPV code (common procurement vocabulary) is a classification of the main subject of procurements unified across the European Union. For more information, visit <http://simap.ted.europa.eu/web/simap/cpv>.

The table, however, shows evident changes in the use of procurement procedures after the reform. More than 16.5 percent of the total procurement value shifted from open auctions to other, less transparent, procedures. This demonstrates the preference of procurement agencies for the new procedures.

We further merge the procurement dataset with additional information about contractors from the official register of economic subjects using unique contractor identifiers. The register includes information about the size of their capital stock, legal form, year of incorporation, number of employees, and—most important—about the traceability of their owners.

In total, procurements transferred to anonymously owned firms were worth CZK 22 billion (approximately US\$1 billion), based on almost 1,200 contracts, which represents 2.6 percent of the total number of contracts. These contracts were awarded to a total of 277 unique anonymously owned firms. The share of these firms is not negligible, as they represent about 20 percent of all joint-stock companies obtaining public procurements.

Table 2 compares anonymously owned joint-stock firms with traceable joint-stock firms. Anonymously owned firms are, on average, smaller, three years newer, and have much smaller capitalization.

III. Empirical Strategy for Detecting Manipulation of Procurement Values

We use two empirical methods for detecting manipulation of procurement contracts. The first is based on the methodology of Chetty et al. (2011) and focuses on repeated cross-sectional density distributions of the anticipated value of procurements. The assumption, which underlies the statistical inference, is that the density distributions of the anticipated value would be smooth if more restrictive tendering procedures were not prohibited above the thresholds. The smoothness assumption might be rather strong in our application, for example in case rounding of the inspected variable were an issue.

Therefore, we offer an alternative strategy, which relaxes the assumption of the smooth counterfactual density distribution by exploiting the timing of the introduction of new thresholds into the procurement system. We assume that the density distribution after the reform would look the same as before the reform, if the reform had not established procurement thresholds. The assumptions behind our strategy are a stable demand for procurements over time, and no intentional delays in procurements until after the reform.⁵ Our empirical strategy is technically only an extension of that in Chetty et al. (2011), thus we start with cross-sectional analysis.

In the first method, we estimate the excess mass of contracts below a certain threshold D using a counterfactual density distribution—what the anticipated value distribution would look like if there were no ban on restrictive tendering above D .

⁵Supportive evidence for the first assumption is presented in Table 1, which shows that the structure of the overall financial volume spent on construction, goods, and services appears to be stable over time. Evidence for the second assumption is discussed in Section IV, in which we show that the extent of manipulation remains stable over time after the reform. Moreover, our evidence rather suggests that there was a slight delay in the impact of the reform, mainly in the construction sector.

TABLE 2—CONTRACTOR CHARACTERISTICS, BY TYPE OF OWNERSHIP

	All companies	Traceable companies	Anonymous companies	Difference
Capital stock (mill. CZK)	334	396	78.2	−317.8
Median year of incorporation	1997	1996	1999	
Number of employees				
0–24 employees	28.77	25.04	44.04	+19.0
25–99 employees	32.46	34.48	24.19	−10.3
100–249 employees	17.36	18.96	10.83	−8.13
250–999 employees	12.83	13.49	10.11	−3.38
1,000 and more	4.61	5.20	2.17	−3.04
Not specified	3.97	2.82	8.66	+5.84

Note: The differences between joint-stock companies with anonymous and traceable owners are tested using one-sample two-group *t*-test in case of the “capital stock”; *k*-sample test of the equality of medians in case of the “median year of incorporation”; and one-sample two-group *z*-tests of the equality of proportions in case of the “Number of employees” categories.

In the first step, we plot the empirical distribution of the anticipated value of procurements in a histogram with *D* re-centered to zero. In the second step, a polynomial is fitted to the histogram excluding the data in a narrow window below the threshold. This means that a polynomial regression of the following form is estimated:

(1)
$$C_j = \sum_{i=0}^q \beta_i \cdot (Z_j)^i + \sum_{i=-R}^0 \gamma_i \cdot \mathbf{1}[Z_j = i] + \varepsilon_j,$$

where *C_j* is the number of procurement contracts in a histogram bin *j*, *Z_j* is the anticipated value of contracts grouped in histogram bin *j*, *q* is the order of the polynomial, and *R* denotes the width of the excluded region below the threshold measured in the number of excluded bins below *D*.⁶

The estimate of the counterfactual distribution is defined as predicted values from (1) omitting the contribution of the dummy variables below the threshold:

(2)
$$\hat{C}_j = \sum_{i=0}^q \hat{\beta}_i \cdot (Z_j)^i.$$

The excess number of contracts that are located below the threshold is⁷

(3)
$$\hat{B}_N = \sum_{j=-R}^0 C_j - \hat{C}_j = \sum_{i=-R}^0 \hat{\gamma}_i.$$

⁶We conduct the analysis for different parametric choices of *q* and *R*. We also estimate the specification where the excluded region is symmetric around the threshold *D*. We comment on the results of this analysis in Section IV.

⁷This calculation would overestimate *B_N* because it does not satisfy the constraint that the area under the counterfactual distribution must equal the area under the empirical distribution. To take this aspect into account, we follow Chetty et al. (2011) and correct the counterfactual distribution above the threshold so that the integration constraint is satisfied.

Finally, we define the empirical estimate of the excess mass below the threshold relative to the average density of the counterfactual anticipated contract value distribution between $-R$ and 0 as

$$(4) \quad \hat{b} = \frac{\hat{B}_N}{\sum_{j=-R}^0 \hat{C}_j / R}.$$

We calculate the standard error for $\hat{b}$ using a parametric bootstrap procedure. We draw from the estimated vector of errors ξ_j in (1) with replacement to generate a new set of counts and apply the above technique to calculate a new estimate $\hat{b}^k$. We define the standard error of $\hat{b}$ as the standard deviation of the distribution of $\hat{b}^k$ s.

The second empirical method incorporates a time dimension into the econometric model. This method relaxes the assumption of a smooth density distribution of the anticipated contract value and assumes that the shape of the density distribution after the 2006 reform would look the same as prior to it.

The estimation again proceeds in two steps. We first plot all the annual histograms of the anticipated value with thresholds re-centered to zero. In the second step, we regress the number of contracts in bin j and time t , denoted as C_{jt} , on an interaction term between an indicator for contracts located in the excluded region below the threshold and indicator for the validity of the 2006 reform (that occurred in time denoted as T). We include in our model a set of fixed effects for histogram bins in which contracts would be located, and annual fixed effects, which are intended to capture the time trend. The econometric model can be formally expressed as follows:

$$(5) \quad C_{jt} = \alpha_j + \alpha_t + \sum_{i=-R}^0 \gamma_i \cdot \mathbf{1}[Z_j = i] \cdot \mathbf{1}[t > T] + \varepsilon_{jt}.$$

The coefficients of interest $\hat{\gamma}_i$, $i \in [1, R]$, represent our estimates of the excess mass of contracts in particular bins of the excluded region below the threshold. We estimate the regressions using Poisson conditional fixed-effects quasi-maximum likelihood (QML). This estimator has several desirable properties, including consistency of the coefficient estimates independently of any assumption on the conditional variance as long as the mean is correctly specified (Wooldridge 1997) and consistency of the standard errors even if the data generating process is not Poisson⁸.

IV. Empirical Analysis of Manipulation Detection

We divide our analysis into three types of procurements: construction, goods, and services. This is because the difficulty of manipulating the value of a procurement may depend on the product type. For example, monitoring an appropriate price for consulting services can be more difficult compared to procurement of IT hardware for which comparison prices are readily available.

⁸The estimation is implemented in STATA with the xtpqml procedure written by T. Simcoe and is available at <http://people.bu.edu/tsimcoe/code/xtpqml.txt>.

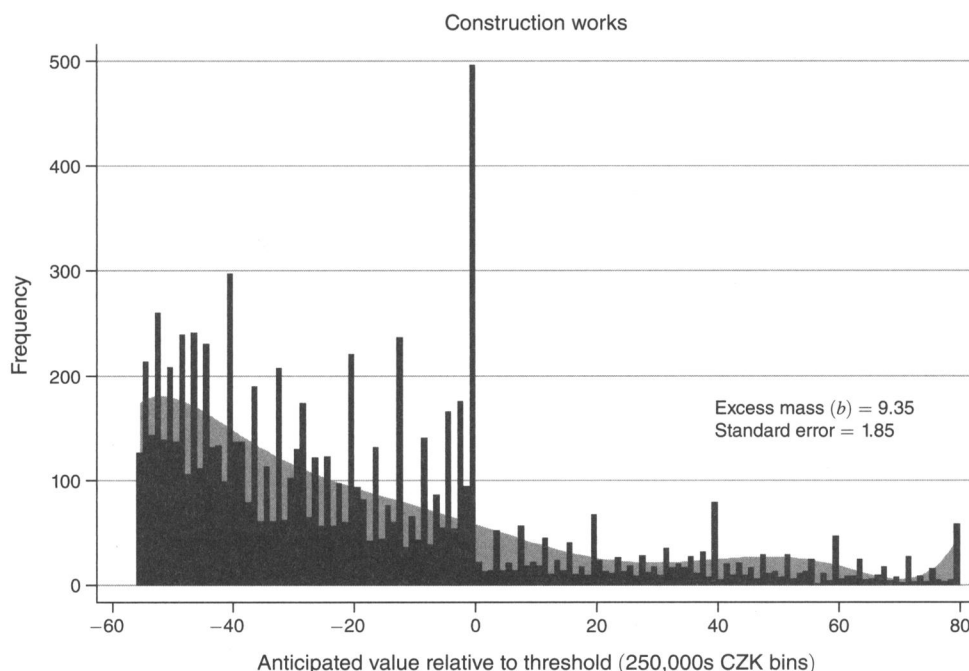


FIGURE 1. ANTICIPATED VALUE DENSITY DISTRIBUTION AROUND THE PROCUREMENT THRESHOLD

Notes: The series shown in bars is a histogram of the anticipated value of construction works, relative to the threshold. The solid distribution beneath the empirical distribution is a seventh-degree polynomial fitted to the empirical distribution, excluding points CZK 750,000 or less below the threshold.

We start with cross-sectional analysis and plot the empirical distribution of the anticipated value of procurements for all construction contracts procured in the Czech Republic after the 2006 procurement reform up to the end of our observation period in 2010 (Figure 1). The contracts are grouped into CZK 250,000 bins ($-14,000,000$ to $-13,750,000$, $-13,750,000$ to $-13,500,000$, etc.) on the re-centered anticipated value variable (the threshold is re-centered to zero).

Figure 1 shows that there is a spike below the simplified negotiations threshold in the (otherwise declining) anticipated value distribution. The solid area beneath the empirical distribution shows the counterfactual density $\{\hat{C}_j\}$ predicted using (1) with a seventh-degree polynomial ($q = 7$) and a window of CZK 750,000 located just below the threshold ($R = 3$). With these parameters, we estimate $b = 9.35$ —the excess mass below the threshold is 935 percent of the average height of the counterfactual distribution within CZK 750,000 below the threshold.^{9, 10}

⁹The results are not qualitatively sensitive to changes in bin size, parametric choice of q or R , as demonstrated in the online Appendix in Table A4, Table A5, and Table A6, respectively. The results are also not sensitive to specifications accounting for specific focal points located within the anticipated value distribution (for example, located at substantial round figures within the distribution).

¹⁰We also estimate the specification where the excluded region is symmetric around the threshold. This exercise increases the estimated magnitude of bunching for goods and services contracts and reduces the estimated bunching for construction contracts. The estimates remain highly statistically significant.

TABLE 3—POLYNOMIAL REGRESSION ESTIMATES OF EXCESS MASS BELOW THE THRESHOLDS

	Construction works	Goods	Services
$\hat{b}$	9.352 [1.850]	1.996 [0.236]	3.027 [0.275]
$\hat{B}_N$	581	282	501
Observations	8,830	5,228	6,357

Notes: $\hat{B}_N$ denotes the estimated excess number of contracts below the thresholds, and $\hat{b}$ denotes the excess mass of contracts relative to the average density at the thresholds. Standard errors are presented in brackets.

The first column of Table 3 presents these results. The standard error associated with our estimate of b is 1.85. The null hypothesis that there is no excess mass at the threshold relative to the counterfactual distribution is rejected with a t -statistic of 5.055.

Using the same methodology, we also find statistically significant evidence of manipulations of the anticipated values of goods and services contracts. The second and third columns of Table 3 summarize these results.¹¹ The estimated excess mass at threshold for goods contracts is 200 percent of the average height of the counterfactual distribution. The estimated excess mass at the threshold for services is 303 percent of the average counterfactual distribution height. The estimated extent of manipulation is thus smaller than for construction.¹²

Using a back-of-the-envelope calculation, we calculate the share of manipulated contracts procured through the simplified negotiations below thresholds. According to our results, the manipulations affect 11 percent of all contracts procured using this non-transparent procedure after July 2006.¹³

The assumption of the smooth counterfactual density distribution can be relaxed by exploiting the timing of the introduction of new thresholds into the procurement legislation. Figure 2 displays the distribution of the anticipated value in each year from 2005–2010 for all construction works. In the subfigures it is clear that the first appearance of bunching appears just a few months after the introduction of the simplified negotiating threshold. In the subfigures, the excess mass of contracts persistently clings very closely to the legislative threshold after the reform and its size remains quantitatively similar. This implies that possible short run effects, for example, intentional delays of procurements until the reform, are not relevant for the estimated extent of manipulation just below the limit.

The annual estimates of discontinuities for all types of the main subject are shown in Table 4. The procurement reform and the first emergence of discontinuities

¹¹ Due to space constraints, we omit figures showing the distributions for goods and services contracts.

¹² We test for the presence of bunching in the five largest subcategories of construction works, including building construction works, engineering works, and construction works for water projects (see Table A7 in the online Appendix). We find evidence of bunching in all of these categories, except for the relatively small category of water projects, for which the estimates are not statistically significant. This finding is different to Coviello, Guglielmo, and Spagnolo (2016), who find no evidence of bunching in the Italian building construction sector.

¹³ We count 1,364 contracts excessively massed below thresholds, while altogether 12,372 procurements were allocated using the simplified negotiations after July 2006.

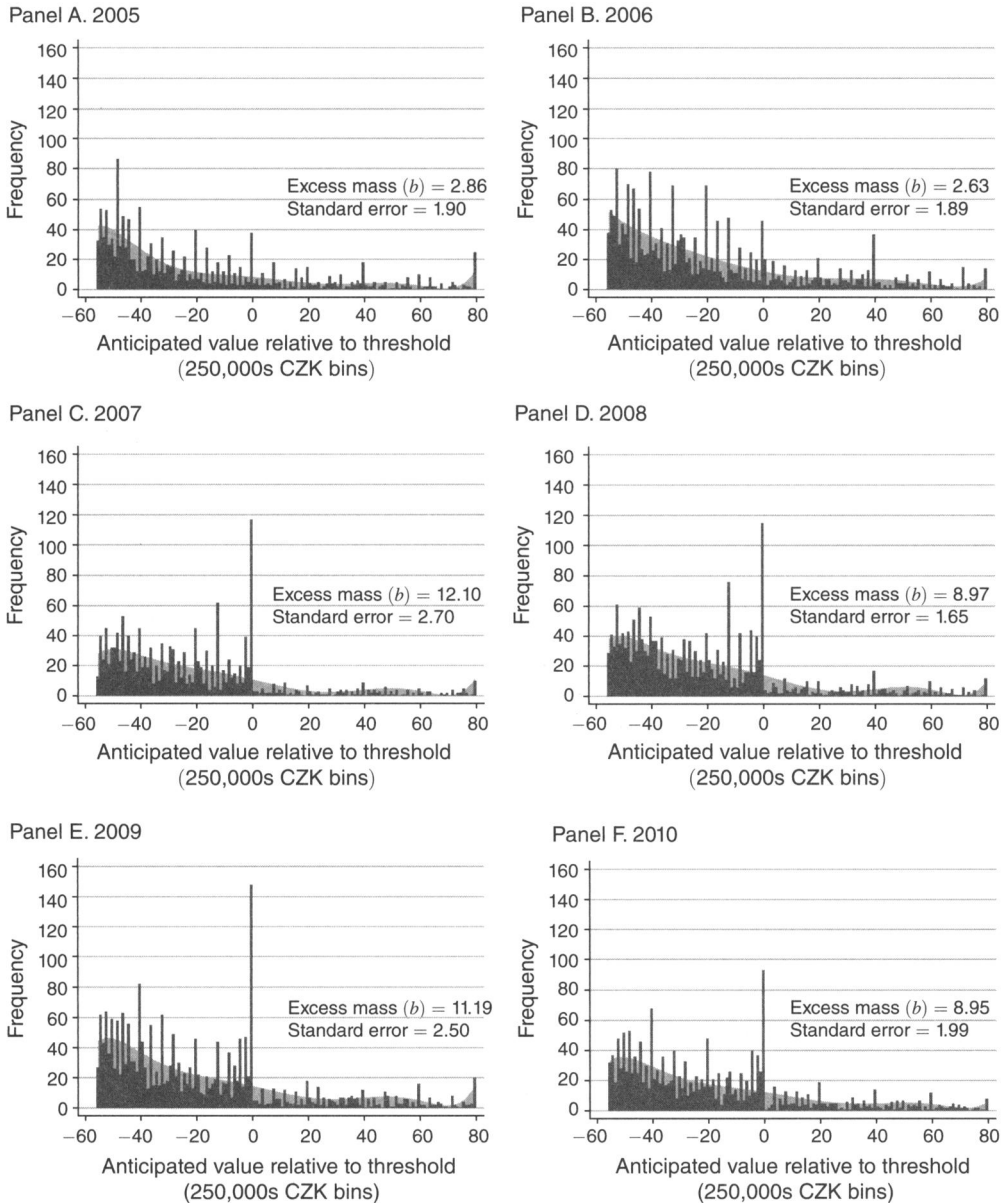


FIGURE 2. ANTICIPATED VALUE DISTRIBUTIONS AROUND PROCUREMENT THRESHOLDS, BY YEAR

Notes: Series shown in bars are histograms of the anticipated value of construction procurements relative to the threshold. Each bar shows the number of observations in CZK 250,000 bins. The solid distributions beneath empirical distributions are seventh-degree polynomials fitted to empirical distributions, excluding points CZK 750,000 or less below the threshold.

coincide perfectly for goods and services contracts. For construction works, the discontinuity can be statistically detected after a six-month period, which is evidence against the hypothesis of intentional waiting for the reform.

The delay in construction can be contrarily explained by the fact the procurement process requires a relatively considerable amount of time, which varies for different

TABLE 4—ESTIMATED EXCESS MASS BELOW THE THRESHOLDS BY YEAR AND MAIN OBJECT

Year	Construction works		Goods		Services	
	Excess mass estimates	SE	Excess mass estimates	SE	Excess mass estimates	SE
2005	2.861	[1.902]	0.410	[0.552]	−0.025	[0.577]
2006	2.628	[1.891]	0.800	[0.294]	1.635	[0.257]
2007	12.100	[2.697]	1.389	[0.427]	3.162	[0.460]
2008	8.965	[1.651]	1.799	[0.494]	2.121	[0.478]
2009	11.190	[2.508]	1.901	[0.522]	2.503	[0.561]
2010	8.954	[1.990]	2.362	[0.360]	2.852	[0.371]

Notes: The table estimates the excess mass of contracts relative to the average density at thresholds. Standard errors are presented in brackets.

types of contracts. Because the new law was passed in July 2006, the authorities may not have procured enough new construction contracts (as opposed to smaller and simpler goods and services contracts) in the last two quarters of 2006 for manipulations to become statistically significant before 2007.

We further continue by quantifying the excess mass using Poisson conditional fixed effects. Table 5 shows the results of estimating (5), which incorporates time into our model, with CZK 250,000 wide histogram bins and CZK 750,000 wide excluded region below the threshold ($R = 3$). Using these parameters, we estimate that the number of contracts in the last bin below the threshold increased after the 2006 reform by 156 percent, 113 percent, and 182 percent for construction works, goods, and services contracts, respectively. The null hypotheses of no manipulation of procurement in these bins were rejected with z -statistics of 24.79, 13.30, and 16.20. The estimated excess bunching is of a smaller but still significant magnitude relative to the cross-sectional analysis.

The results presented in Table 5 are our preferred estimates of bunching, as they take into account the pre-reform distribution of procurement values, which can be influenced, for example, by rounding of procurement values. This can be confirmed by a visual inspection of Figure 2, which shows that the distribution before the reform exhibits some accumulation of contracts at approximately every CZK 1 million.

Results from both cross-sectional and policy change empirical strategies suggest that manipulation is more pronounced for construction and services, while it is less pronounced, but still significant, for goods. Our explanation is that the subject of contracts is much more difficult to specify in procurements on construction and services and the scope for manipulation is much broader. As a result, procurements on construction and services are generally more difficult to control and provide more opportunities for manipulation relative to procurements of goods.

Robustness Checks.—We provide alternative approaches to detection of manipulation and analysis of robustness. In particular, we test whether procurements cluster at inflationary adjusted thresholds and use the alternative density test described by McCrary (2008) to test for procurements bunching.

Does Contract Bunching Follow Inflation or Does It Cling to Thresholds?—One could still conjecture that at the time of the procurement reform a change occurred in government orders for projects that were worth approximately the same value as

TABLE 5—ESTIMATES OF EXCESS MASS BELOW THE THRESHOLDS
USING A FIXED-EFFECTS STRATEGY

	Construction	Goods	Services
$\hat{\gamma}_{-1}$	0.942 [0.038]	0.758 [0.057]	1.037 [0.064]
$\hat{\gamma}_{-2}$	1.478 [0.038]	0.295 [0.057]	0.006 [0.064]
$\hat{\gamma}_{-3}$	1.205 [0.038]	0.571 [0.057]	0.188 [0.064]
Histogram bin fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	816	990	996

Notes: Coefficient estimates are interpreted as $(\exp(\hat{\gamma}_i) - 1) \times 100$ percentage change. Robust standard errors, clustered at the histogram bin level, are presented in brackets.

the procurement threshold. Such a change in government needs would have brought about a disproportionate representation of projects beneath the threshold even in the absence of any manipulation. However, one would then expect that the spike in the anticipated value density distribution would shift with inflation over time.

In Figure 3 we consider the period from 2007 to 2010, during which the simplified negotiations threshold for construction works declined in real terms. Noting that the excess mass was located at the negotiations threshold in 2007, the figure shows two possibilities for its location in 2010: the 2010 threshold and the 2007 threshold adjusted for inflation in the construction industry.

Figure 3 shows that the excess mass clearly clings to the 2010 threshold rather than following inflation. The procurement threshold is therefore more important for contract bunching than specific government needs.

Alternative Density Discontinuity Test.—We apply McCrary’s (2008) density test in order to provide an alternative test of contract bunching. The test consists of an extension of the local linear density estimator from Cheng, Fan, and Marron (1997) and is particularly useful in applications where a discontinuous density is itself the object of interest. In a practical sense, the test is implemented as a Wald test of the null hypothesis that there is no discontinuity at threshold D . Table 6 presents the log discontinuities estimated using McCrary’s (2008) local linear regressions along with the simulated standard errors.^{14, 15} The results in the table strongly suggest that the density function of anticipated value of procurements is discontinuous at the threshold for simplified negotiations. The estimates indicate that contracts within hundreds of thousands of CZK of the threshold are much more likely to be procured

¹⁴To apply the local linear density estimator, we select the bandwidth of $h = 2,000,000$ and the bin size of $w = 250,000$ subjectively after using an automatic procedure. Pagan and Ullah (1999), and Deaton (1997) point out the effectiveness of subjective bandwidth choice. The automatic procedure would select a bin size of $w = 191,313$ and a bandwidth of $h = 4,749,168$.

¹⁵We follow Horowitz (2001), Hall (1992), and McCrary (2008) and, when estimating the standard error, we under-smooth the local linear estimator by choosing a half bandwidth with respect to the reference bandwidth. The cited authors recommend this procedure in order to reduce the bias associated with a bandwidth which minimizes the asymptotic mean square error.

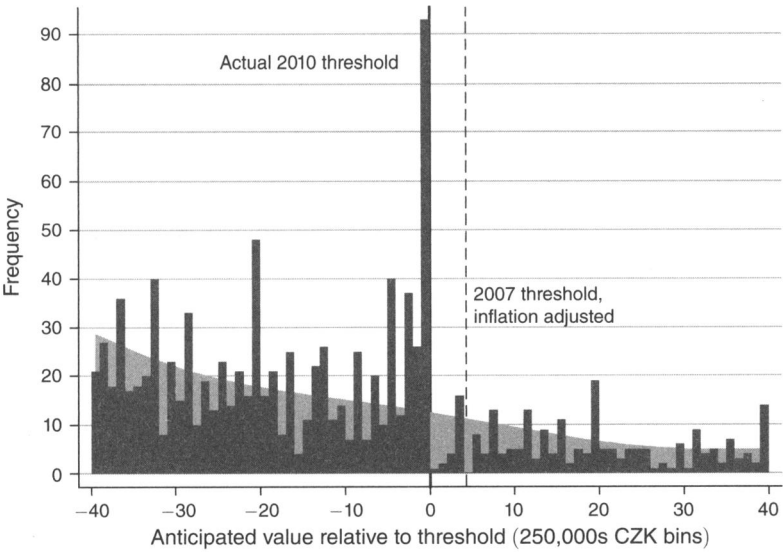


FIGURE 3. DISTINGUISHING THRESHOLD IN PUBLIC PROCUREMENT FROM INFLATION

Notes: The location of the threshold in 2010 is marked with a solid line. The dashed line shows the level of the 2007 threshold adjusted for inflation in the Czech construction industry.

TABLE 6—LOG DENSITY DISCONTINUITY ESTIMATES

	Construction works	Goods	Services
$\hat{\theta}$	−3.291 [0.243]	−0.457 [0.104]	−0.801 [0.079]
Observations	9,067	6,869	8,518

Notes: The table presents the log estimates of discontinuity in the density of the anticipated value. Estimates were obtained using a local linear density estimator proposed by McCrary (2008). Simulated standard errors are presented in brackets.

below the threshold than above it. These results affirm the robustness of our prior analysis of contract bunching.

V. Manipulations and Sorting of Firms below Thresholds

The results presented so far provide robust evidence of manipulations of anticipated value, especially for construction works and services. In this section, we illustrate how the introduction of procurement thresholds affects sorting of specific contractors below thresholds and how procurement characteristics change in their proximity.

In particular, we focus on the allocation of procurements to firms which allow hidden owners. According to the previous literature, firms similar to those with anonymous owners can serve as a means of transferring rent from procurement (Mironov and Zhuravskaya 2016).

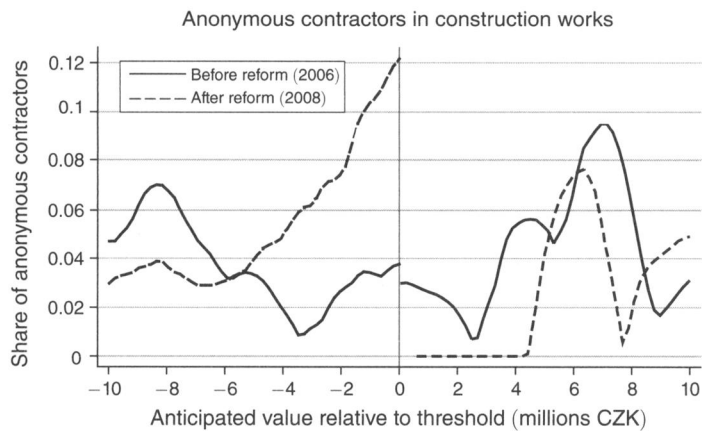


FIGURE 4. SHARE OF CONSTRUCTION CONTRACTS AWARDED TO FIRMS WITH ANONYMOUS OWNERS, BY YEAR

Figure 4 illustrates the access of anonymously owned firms to procurements in the proximity of the threshold for construction works. The figure shows the share of procurements awarded to anonymously owned joint-stock companies (out of all joint-stock companies) in two periods: before and after the 2006 reform that established the thresholds. We use kernel-weighted local polynomial smoothers and CZK 1 million bandwidths on either side of the threshold to plot the figure.

Figure 4 clearly indicates that after the reform, contracts just below the threshold were three times more likely to be allocated to firms with anonymous owners than prior to the reform.

We also inspect the access of firms with anonymous owners to contracts in a regression framework for all types of procurements: construction works, goods, and services. We apply a similar methodology to that leading to equation (5). For that purpose, we create a binary variable for whether the contract was awarded to a company with anonymous owners or not. We regress this measure of access to contracts on a variable that captures the interaction between contracts located just below thresholds and the legal force of the 2006 reform. Our specification also includes annual fixed effects, fixed effects for histogram bins in which contracts are located, and dummies for all types of procurement procedures. Finally, the specification controls for the content of procurements as it contains dummies for procurement CPV codes that systematically characterize the detailed procurement subject.¹⁶

This regression is a statistical model to detect sorting of anonymously owned firms below the discretionary thresholds where we estimate a significant bunching of procurement values. The model is based on the comparison of empirical distributions of anonymously owned firms before and after the introduction of the thresholds, similarly as in the case of procurement values. We therefore formally test

¹⁶Technically, we control for the content of procurement at the level of the first three digits of the detailed CPV code. Our specification includes dummies for the most prevalent CPV code groups within each type of procurement—this materializes in 6 CPV code dummies for construction contracts, 16 dummies for services, and 15 dummies for goods.

TABLE 7—SORTING OF CONTRACTORS BELOW THRESHOLDS

Outcome variable: Indicator that contractor is anonymously owned						
	Construction works		Services		Goods	
Contracts in bins just below $D \times 2006$ reform	0.027 [0.012]	0.029 [0.012]	0.011 [0.006]	0.013 [0.008]	−0.006 [0.015]	−0.004 [0.015]
Histogram bin fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Procurement procedure dummies	No	Yes	No	Yes	No	Yes
Procurement subject (CPV code) dummies	No	Yes	No	Yes	No	Yes
R^2	0.01	0.01	0.06	0.11	0.02	0.04
Observations	11,863	11,585	7,117	7,016	7,494	7,398

Notes: Estimates multiplied by 100 can be interpreted as percentage point change. Robust standard errors, clustered at the histogram bin level, are presented in brackets.

whether the sorting of anonymously owned firms appears just below the threshold simultaneously with the bunching of procurement values. The regressions are estimated using the linear probability model (LPM), with standard errors clustered at the histogram bin level.

Table 7 presents the results of this estimation and supports the findings presented in Figure 4 for the construction sector.

In particular, Table 7 suggests that after the 2006 reform, the conditional probability that a contractor just below the threshold is anonymously owned increased by approximately 3 percentage points in construction. This corresponds to an approximate threefold increase in the probability of officials awarding a contract to an anonymously owned firm. The size of the increase is comparable to Figure 4, even though the regression sample now includes all contractors, not only joint-stock companies. This result is significant at the 5 percent level. For services, we find over one percentage point increase in the prevalence of anonymously owned firms below the threshold, which is significant at the 10 percent level. In the case of goods, we estimate a non-significant change in the prevalence of anonymously owned firms just below thresholds.

Putting our evidence together, we observe greater sorting of anonymously owned firms in construction and services, where, simultaneously, the bunching of procurement values is also more prevalent. The introduction of thresholds therefore drives both the bunching of procurements and, at the same time, the sorting of anonymously owned firms. It should be noted that the detected manipulations may have also led to other changes in the characteristics of procurements after the introduction of thresholds. These results therefore provide a concrete illustration of how economic behavior may violate assumptions behind regression discontinuity designs in settings with high incentives to manipulate anticipated values.

Nevertheless, we consider the emergence of sorting of anonymously owned firms below thresholds after the reform as the revealed preference of procurement agencies for hidden owners. The selection of an anonymously owned contractor may or may not be associated with other procurement characteristics; but in either case, we consider it as a threat to the transparency and fairness of the procurement process.

TABLE 8—ANONYMOUSLY OWNED CONTRACTORS AND PROCUREMENT PRICES

Outcome variable: Difference between the contract price and anticipated value of procurement (in percent of anticipated value)						
	Construction works		Services		Goods	
Contracts in bins just below $D \times$ anonymous firm	0.082 [0.034]	0.089 [0.029]	0.089 [0.026]	0.059 [0.029]	−0.063 [0.050]	−0.055 [0.053]
Anonymous firm	−0.016 [0.018]	−0.014 [0.015]	−0.000 [0.025]	0.018 [0.024]	0.042 [0.038]	0.033 [0.044]
Histogram bin fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Procurement procedure dummies	No	Yes	No	Yes	No	Yes
Procurement subject (CPV code) dummies	No	Yes	No	Yes	No	Yes
R^2	0.01	0.06	0.04	0.06	0.01	0.01
Observations	8,241	7,976	6,051	5,953	6,024	5,931

Notes: Estimates multiplied by 100 can be interpreted as percentage point change. Robust standard errors, clustered at the histogram bin level, are presented in brackets.

We further ask whether sorting of anonymously owned firms below the thresholds is associated with changes in procurement prices. Table 8 documents that procurements of anonymously owned firms have higher contract prices compared to procurements with the same anticipated value below the thresholds, but awarded to firms with traceable owners. This illustrates that procurements awarded to firms with anonymous owners can be more costly compared to procurements allocated to firms with traceable owners. However, as previously mentioned, procurements allocated to anonymously owned firms may have different underlying value compared to procurements granted to firms with traceable owners. One reason may be that reputable traceable firms may not be willing to participate in competitions together with anonymously owned firms, which may generate differences in the content of procurements. To mitigate this bias, the regressions include CPV code dummies, which to some extent control for the procurement subject.

Furthermore, the regressions include histogram bin fixed effects to account for changes in the relative difference between the contract price and anticipated value of procurements across the whole distribution of the anticipated values. Nevertheless, if contracts awarded just below the thresholds to anonymously owned firms still have on average higher underlying value and are properly delivered, then the estimated higher contract price may not be interpreted as an increased cost in the procurement.

The coefficient of interest in Table 8 is the interaction of the indicators for anonymously owned firms and contracts located just below the threshold (CZK 750,000 below or less). Using the post-2006 sample, we estimate an 8.9 percentage point higher price of construction contracts awarded to anonymously owned firms just below the thresholds. The estimate is significant at the 1 percent level. Similarly, we find significant association between anonymously owned firms and procurement prices among manipulated service contracts. The estimated effect is

5.9 percentage points. For goods where the sorting of anonymously owned contractors is not present, there is no association between price and allocation of procurements to firms with untraceable owners.

VI. Conclusion

In this paper, we detect and quantify manipulation of procurement contracts using the example of the Czech public procurement system. An incentive for officials to manipulate the anticipated value of procurements was created by a policy reform that introduced new discretionary thresholds, below which the procurement agencies were given autonomy to preselect any five contractors for the bidding process.

We quantify the extent of manipulation using two empirical strategies. In the first, we use the methodology presented in Chetty et al. (2011) to estimate the cross-sectional counterfactual distributions of procurement value. The second strategy is our extension, which employs the distributions of procurement value before the reform as counterfactual.

Using both methods, we find a substantial impact of the policy reform on the extent of manipulation, even though our method yields lower estimates of bunching. We further find that manipulation is simultaneously related to a threefold increase in the probability that construction procurements are awarded to firms with anonymous owners. We consider this finding as worrying, because the anonymity of company owners allows for hiding rent-seeking behavior as it can conceal the identity of agents who are procurement officials and at the same time stakeholders in anonymously owned companies. Although we estimate that procurements just below the threshold are associated with higher contract prices if they are allocated to anonymously owned firms rather than to firms with traceable owners, we cannot provide evidence of overall losses in welfare due to data limitations.

To calculate the total costs of manipulation, one would need to obtain information about the quality of the provided procurements, details about contract renegotiations, and information about the administrative costs of contract-allocation procedures. Although anonymously owned suppliers are most often selected using a simplified procedure, which may save some administrative costs, at the same time, anonymous ownership restricts the transparency of the procurement process. This may not only hide conflicts of interest, but can also distort competition in procurement markets and discourage other potential suppliers from participation. Such effects may be difficult to quantify and may not become evident immediately. On the positive side, greater discretion on the part of procurement officials can also serve to speed up the procurement process by allowing preselection of incumbent suppliers (Coviello, Guglielmo, and Spagnolo 2016; Bobilev et al. 2015). Elimination of firms with poor past performance can help officials to ensure the quality of procurements and limit contract renegotiations (Bannerjee and Duflo 2000, Calzolari and Spagnolo 2009). Although some procurements may have thus been manipulated with the intention to improve welfare, the total effect is difficult to quantify.

Should thus procurement officials be allowed to preselect potential contractors below thresholds with a high degree of autonomy? In general, if there is misalignment between the preferences of officials and those of society, the optimal

delegation literature (e.g., Alonso and Matouschek 2008) advises calling for stricter rules and external controls. In the spirit of Holmstrom and Milgrom (1991), it might also be optimal to strip procurement officials of the discretion to preselect potential contractors autonomously, and rather leave them with the option of including past performance indicators among bid evaluation criteria. This policy may be specifically relevant for countries with weak institutions and low accountability of public officials.

Finally, our paper provides controlling bodies with a new tool for analyzing manipulation in procurement competition. We illustrate the scope of manipulation using the case of the Czech Republic, but many other countries regulate their procurements using similar thresholds, although with different changes in discretion at thresholds. This includes, for example, public administrations in the European Union, as EU directives on public procurement cover only tenders expected to be worth more than a given threshold. Each country in the European Union can create its own procurement thresholds, provided that the thresholds are located below those determined by EU directives. Similarly, the Federal Acquisition Regulations in the United States permit simplified acquisition procedures only for supplies and services, the aggregate amount of which does not exceed the simplified acquisition threshold. Given the often limited resources of controlling bodies, we suggest that it may be cost-efficient to use our methodology to find the extent of manipulation and further test the extent to which it has consequences on the price of procurements, undesirable selection of winners, and further procurement outcomes.

APPENDIX

TABLE A1—PROCUREMENT THRESHOLDS FOR SIMPLIFIED NEGOTIATING PROCEDURE (*in thousands of CZK*)

	2005	2006	2007	2008	2009	2010
Goods, services						
National bodies	N/A	4,290	4,290	3,782	3,782	3,236
Regional bodies	N/A	6,607	6,607	5,857	5,857	4,997
Construction works	N/A	20,000	20,000	20,000	20,000	20,000

Notes: Table A1 shows the annual procurement thresholds for simplified negotiations by the main object of procurements and the type of contract-awarding agency (in thousands of CZK). Thresholds for simplified negotiations determine the level of discretion of public officials to invite suppliers of their choice. Thresholds also restrict entry of bidders and determine the overall transparency of the contract-awarding process. Simplified negotiation thresholds were introduced on July 1, 2006.

Source: Consolidated text of act no. 137/2006 Coll. on Public Contracts

TABLE A2—PROCUREMENT PROCEDURES

Open procedure	In the open procedure, a procurement agency starts the procurement process by publishing complete project specifications, including prequalification criteria for potential suppliers, anticipated project value, and bid evaluation criteria. A deadline for bid submission is scheduled, and any firm may submit a bid. Upon the deadline, the procuring agency reviews whether the firms have fulfilled the prequalification criteria. From among the qualified firms, the agency selects the most favorable bid according to the predefined evaluation criteria. This is the default procurement procedure and can be used both below and above procurement thresholds. The requirement to prepare complete project specifications before the start of the procurement process and the rule that any firm can participate in the bidding limits the flexibility of the procurement process and the discretion of the procurement agency to eliminate undesired firms from competition.
Simplified negotiating procedure	In the simplified negotiating procedure, a procurement agency prepares complete project specifications, including anticipated project value and a set of bid evaluation criteria, and sends it to no fewer than five potential suppliers with a call to submit bids. The firm submitting the lowest bid is selected as the supplier. This procedure may be used only when the anticipated value of a procurement does not exceed the procurement threshold. The selection of firms to be invited is at the full discretion of the procurement agency. This procedure gives much more flexibility and discretion to the agencies compared to the open procedure; however, this is at the cost of less equal treatment of potential suppliers and less transparency.
Other procurement procedures	There are several other procurement procedures that can be used irrespective of procurement thresholds: for example, a restricted procedure or a negotiating procedure without prior public notice. As shown in Figure A1 in the online Appendix, these procedures are not affected by procurement thresholds, as we do not observe any bunching at the thresholds with these procedures.

REFERENCES

Alonso, Ricardo, and Niko Matouschek. 2008. "Optimal Delegation." *Review of Economic Studies* 76 (1): 259–93.

Bandiera, Oriana, Andrea Prat, and Tommaso Valletti. 2009. "Active and Passive Waste in Government Spending: Evidence from a Policy Experiment." *American Economic Review* 99 (4): 1278–1308.

Banerjee, Abhijit V., and Esther Duflo. 2000. "Reputation Effects and the Limits of Contracting: A Study of the Indian Software Industry." *Quarterly Journal of Economics* 115 (3): 989–1017.

Banfield, Edward C. 1975. "Corruption as a Feature of Governmental Organization." *Journal of Law and Economics* 18 (3): 587–605.

Bobilev, Roman, Andrea Guglielmo, Elena Paltseva, and Giancarlo Spagnolo. 2015. "Public Procurement Thresholds and Data in Sweden." Konkurrensverket (KKV) Report 2015:3.

Bulow, Jeremy, and Paul Klemperer. 1996. "Auctions Versus Negotiations." *American Economic Review* 86 (1): 180–94.

Calzolari, Giacomo, and Giancarlo Spagnolo. 2009. "Relational Contracts and Competitive Screening." Centre for Economic Policy Research (CEPR) Discussion Paper 7434.

Cheng, Ming-Yen, Jianqing Fan, and James S. Marron. 1997. "On Automatic Boundary Corrections." *Annals of Statistics* 25 (4): 1691–1708.

Chetty, Raj, John N. Friedman, Tore Olsen, and Luigi Pistaferri. 2011. "Adjustment Costs, Firm Responses, and Micro vs. Macro Labor Supply Elasticities: Evidence from Danish Tax Records." *Quarterly Journal of Economics* 126 (2): 749–804.

Coviello, Decio, Andrea Guglielmo, and Giancarlo Spagnolo. 2016. "The Effect of Discretion on Procurement Performance." Centre for Economic Policy Research (CEPR) Discussion Paper 11286.

- Coviello, Decio, and Mario Mariniello. 2014. "Publicity requirements in public procurement: Evidence from a regression discontinuity design." *Journal of Public Economics* 109: 76–100.
- Deaton, Angus. 1997. *The Analysis of Household Surveys: A Microeconomic Approach to Development Policy*. Washington, DC: World Bank.
- Di Tella, Rafael, and Ernesto Schargrotsky. 2003. "The Role of Wages and Auditing during a Crack-down on Corruption in the City of Buenos Aires." *Journal of Law and Economics* 46 (1): 269–92.
- Economist. 2013. "Czech finance: Bearer beware." May 19. <http://www.economist.com/blogs/easternapproaches/2013/05/czech-finance>.
- Europe Economics. 2006. *Evaluation of Public Procurement Directives*. European Commission. London, September.
- European Commission. 2011. *Public Procurement in Europe: Cost and Effectiveness*. European Commission, PwC, and London Economics and Ecorys. London, March.
- Gordon, Richard K. 2009. *Laundering the Proceeds of Public Sector Corruption*. Washington, DC: World Bank.
- GovConChannel. 2015. "Got Your Fingers Crossed Small Federal Contractors?" July 25. <https://govconchannel.com/2015/07/25/efficiency-helps-smallbiz/>.
- Hall, Peter. 1992. "Effect of Bias Estimation on Coverage Accuracy of Bootstrap Confidence Intervals for a Probability Density." *Annals of Statistics* 20 (2): 675–94.
- Helper, Susan, and Rebecca Henderson. 2014. "Management Practices, Relational Contracts, and the Decline of General Motors." *Journal of Economic Perspectives* 28 (1): 49–72.
- Holmstrom, Bengt, and Paul Milgrom. 1991. "Multitask Principal-Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design." *Journal of Law, Economics, and Organization* 7: 24–52.
- Horowitz, Joel L. 2001. "The Bootstrap." In *The Handbook of Econometrics*, Vol. 5, edited by James J. Heckman and Edward Leamer, 3159–3228. Amsterdam: North-Holland.
- Kelman, Steven. 1990. *Procurement and Public Management: The Fear of Discretion and the Quality of Government Performance*. Washington, DC: AEI Press.
- McCrary, Justin. 2008. "Manipulation of the running variable in the regression discontinuity design: A density test." *Journal of Econometrics* 142 (2): 698–714.
- Mironov, Maxim, and Ekaterina Zhuravskaya. 2016. "Corruption in Procurement and the Political Cycle in Tunneling: Evidence from Financial Transactions Data." *American Economic Journal: Economic Policy* 8 (2): 287–321.
- Office for Drug Control and Crime Prevention. 1998. *Financial havens, banking secrecy and money-laundering*. United Nations. New York, June.
- Organisation for Economic Co-operation and Development (OECD). 2013. *Government at a Glance 2013*. Paris: OECD Publishing.
- Pagan, Adrian, and Aman Ullah. 1999. *Nonparametric Econometrics*. Cambridge: Cambridge University Press.
- Palguta, Ján, and Filip Pertold. 2017. "Manipulation of Procurement Contracts: Evidence from the Introduction of Discretionary Thresholds: Dataset." *American Economic Journal: Economic Policy*. <https://doi.org/10.1257/pol.20150511>.
- Pavel, Jan. 2013. *Veřejné zakázky a efektivnost*. Praha, Czech Republic: Ekopress.
- Saez, Emmanuel. 2010. "Do Taxpayers Bunch at Kink Points?" *American Economic Journal: Economic Policy* 2 (3): 180–212.
- Schwab, Klaus. 2011. *The Global Competitiveness Report 2011–2012*. Geneva: World Economic Forum.
- Šedová, Jindřiška. 2014. "Transparency of Shareholders in the Czech Republic." *Acta Universitatis Agriculturae et Silviculturae Mendelianae Brunensis* 62 (2): 415–26.
- Sharman, J. C. 2010. "Shopping for Anonymous Shell Companies: An Audit Study of Anonymity and Crime in the International Financial System." *Journal of Economic Perspectives* 24 (4): 127–40.
- Shleifer, Andrei, and Robert W. Vishny. 1993. "Corruption." *Quarterly Journal of Economics* 108 (3): 599–617.
- Shleifer, Andrei, and Robert W. Vishny. 1994. "Politicians and Firms." *Quarterly Journal of Economics* 109 (4): 995–1025.
- Spagnolo, Giancarlo. 2012. "Reputation, competition, and entry in procurement." *International Journal of Industrial Organization* 30 (3): 291–96.
- Transparency International. 2009. *Efektivnost fungování kontrolních systémů veřejných zakázek v České republice*. Prague: Transparency International.
- Transparency International. 2012. *Corruption Perceptions Index 2012*. Berlin: Transparency International.
- Urquiola, Miguel, and Eric Verhoogen. 2009. "Class-Size Caps, Sorting, and the Regression-Discontinuity Design." *American Economic Review* 99 (1): 179–215.
- Wooldridge, Jeffrey M. 1997. "Multiplicative Panel Data Models without the Strict Exogeneity Assumption." *Econometric Theory* 13 (5): 667–78.